

AI / ML / Deep Learning / GenAI Interview Questions with Answers

Python • Machine Learning • Deep Learning • LangChain • LangGraph

1. What is Python and why is it used in AI/ML?

Python is a high-level programming language known for simplicity and rich libraries. It is widely used in AI/ML because of frameworks like NumPy, Pandas, scikit-learn, TensorFlow, and PyTorch.

2. Key features of Python

Easy syntax, interpreted language, large community, extensive libraries, cross-platform compatibility, and strong support for data science and machine learning.

3. Difference between list and tuple

Lists are mutable (can change values) while tuples are immutable (cannot change after creation).

4. What is a set in Python?

A set is an unordered collection of unique elements used to remove duplicates and perform set operations.

5. What is a dictionary?

A dictionary stores data in key-value pairs and allows fast lookup using keys.

6. What is list comprehension?

A concise way to create lists using a single line expression such as `[x*x for x in range(5)]`.

7. What is a lambda function?

A small anonymous function defined using the lambda keyword.

8. What is a generator?

A generator yields values one at a time using the yield keyword instead of returning all values at once.

9. Iterator vs Generator

An iterator is an object with `__iter__` and `__next__` methods. A generator is a simpler way to create iterators.

10. What is exception handling?

Handling runtime errors using try, except, finally blocks.

11. What is Machine Learning?

Machine Learning is a subset of AI where systems learn patterns from data to make predictions or decisions.

12. Types of Machine Learning

Supervised learning, Unsupervised learning, and Reinforcement learning.

13. Supervised learning

A learning approach where models are trained on labeled data.

14. Unsupervised learning

A learning approach where models find patterns in unlabeled data.

15. Reinforcement learning

A learning method where an agent learns by interacting with an environment and receiving rewards.

16. Classification vs Regression

Classification predicts categories while regression predicts continuous numeric values.

17. What is overfitting?

When a model performs well on training data but poorly on unseen data.

18. What is underfitting?

When a model is too simple to capture patterns in the data.

19. Bias vs Variance

Bias is error from overly simple assumptions; variance is error from sensitivity to training data.

20. Train-test split

Dividing dataset into training data and testing data to evaluate model performance.

21. Cross validation

A method to evaluate models by splitting data into multiple folds.

22. Feature engineering

Creating useful input features from raw data to improve model performance.

23. Normalization vs Standardization

Normalization scales data between 0 and 1, while standardization scales data around mean 0 and standard deviation 1.

24. Linear Regression

A supervised algorithm used to predict continuous values using a linear relationship.

25. Logistic Regression

A classification algorithm used to predict probability of classes.

26. Decision Tree

A model that splits data into branches based on conditions.

27. Random Forest

An ensemble method combining multiple decision trees for better accuracy.

28. K-Nearest Neighbors

Predicts a value based on the closest data points.

29. Support Vector Machine

A model that finds the optimal hyperplane separating classes.

30. K-Means Clustering

An unsupervised algorithm that groups data into K clusters.

31. PCA

Principal Component Analysis reduces dimensionality while preserving variance.

32. Precision

Percentage of true positive predictions among predicted positives.

33. Recall

Percentage of actual positives correctly predicted.

34. F1 Score

Harmonic mean of precision and recall.

35. ROC-AUC

A metric evaluating classification model performance across thresholds.

36. Deep Learning

A subset of ML using neural networks with multiple layers.

37. Neural Network

A computational model inspired by the brain consisting of layers of neurons.

38. Activation function

A function that introduces nonlinearity in neural networks.

39. ReLU

A popular activation function defined as $\max(0, x)$.

40. Sigmoid

Activation function mapping values between 0 and 1.

41. Forward propagation

Passing inputs through the network to generate outputs.

42. Backpropagation

Updating network weights using gradients from the loss function.

43. Loss function

A measure of difference between predicted and actual values.

44. Gradient descent

Optimization algorithm that updates parameters to minimize loss.

45. Vanishing gradient problem

When gradients become too small during backpropagation.

46. Dropout

A regularization technique that randomly disables neurons during training.

47. Batch normalization

Normalizes layer inputs to speed up training.

48. CNN

A neural network architecture used for image processing.

49. Pooling

A technique to reduce feature map size.

50. RNN

A neural network designed for sequential data.

51. LSTM

A special RNN capable of learning long-term dependencies.

52. Transfer learning

Using a pre-trained model for a new task.

53. What is NumPy?

A Python library for numerical computing using arrays.

54. What is Pandas?

A library used for data manipulation and analysis.

55. DataFrame

A table-like structure in Pandas for storing data.

56. Handling missing data

Techniques include removing rows or filling values using mean, median, or interpolation.

57. Outliers

Extreme data points significantly different from other observations.

58. Data preprocessing

Cleaning and transforming raw data before modeling.

59. Generative AI

AI systems capable of generating new content such as text or images.

60. LLM

A Large Language Model trained on huge text datasets to generate human-like responses.

61. Prompt engineering

Designing prompts to guide LLM outputs effectively.

62. Tokenization

Breaking text into smaller units called tokens.

63. Embeddings

Vector representations of text capturing semantic meaning.

64. Hallucination in LLMs

When models generate incorrect or fabricated information.

65. RAG

Retrieval Augmented Generation combines LLM generation with external knowledge retrieval.

66. Vector database

A database designed to store embeddings for similarity search.

67. Temperature in LLMs

A parameter controlling randomness of generated output.

68. What is LangChain?

LangChain is a framework used to build applications powered by large language models.

69. LangChain components

Prompt templates, chains, memory, tools, agents, and retrievers.

70. Chain in LangChain

A sequence of steps connecting prompts, LLM calls, and outputs.

71. Memory in LangChain

Stores conversation context to enable multi-turn interactions.

72. Agents in LangChain

Agents allow LLMs to decide which tools to use to complete tasks.

73. Tools in LangChain

External APIs or functions that agents can call.

74. What is LangGraph?

LangGraph is a framework for building stateful multi-step workflows for LLM applications.

75. Node in LangGraph

A node represents a step in a workflow such as an LLM call or tool execution.

76. Edge in LangGraph

Edges connect nodes and define execution flow.

77. State in LangGraph

Shared data that moves between nodes during execution.

78. Conditional edge

A workflow branch that executes based on conditions.

79. Multi-agent systems

Systems where multiple AI agents collaborate to complete tasks.

80. AI deployment

Process of integrating trained models into production applications.

81. Model monitoring

Tracking model performance after deployment.

82. Feature store

A centralized repository for storing and managing ML features.

83. MLOps

Practices combining machine learning and DevOps to automate model lifecycle.

84. Explainability in AI

Methods used to understand how AI models make decisions.

85. Ethical AI

Ensuring fairness, transparency, and responsibility in AI systems.

86. Fine-tuning LLMs

Training a pre-trained LLM further on domain-specific data.

87. Prompt chaining

Using multiple prompts sequentially to solve complex tasks.

88. Agent workflow

An architecture where LLMs decide actions dynamically.

89. Streaming responses

Sending generated tokens gradually instead of waiting for full output.

90. Context window

The maximum number of tokens an LLM can process at once.

91. Model inference

The process of using a trained model to generate predictions.

92. *AI system design*

Designing scalable AI pipelines including data ingestion, model training, and deployment.